

JEEVAN SAI REDDY CHOPPA

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SUMMARY

MS Data Science candidate at the University of Maryland (GPA: 3.86/4.0) specializing in **NLP**, **retrieval-augmented generation**, and **financial time series modeling**. Experienced in building end-to-end ML pipelines - spanning data ingestion, model orchestration, hybrid search infrastructure, and automated evaluation - with a focus on production-ready, measurable outcomes.

EDUCATION

University of Maryland, College Park

Master of Science in Data Science | GPA: 3.86/4.0

College Park, Maryland, USA

May 2026

- **Relevant Coursework:** Principles of Machine Learning, Natural Language Processing, Deep Learning, Time Series Analysis, Big Data Systems, Cloud Computing, Probability & Statistics

Sree Vidyanikethan Engineering College

Bachelor of Technology in Electronics & Communication Engineering | GPA: 3.67/4.0

Tirupati, Andhra Pradesh, India

May 2024

- **Relevant Coursework:** Machine Learning, Discrete Mathematics, Stochastic Probability, Multivariable Calculus, Control Systems

TECHNICAL SKILLS

- **Languages:** Python, SQL, MATLAB
- **ML & NLP:** PyTorch, Scikit-learn, LLM Integration, Deployment & Prompting, Hugging Face Transformers, LangChain, RAGAS, BioClinicalBERT, BioMistral
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, PySpark, Streamlit
- **Time Series & Econometrics:** ARIMA, GARCH, Prophet, Statsmodels
- **Infrastructure & Tools:** Docker, Weaviate, Ollama, AWS, Git/GitHub, MySQL, PostgreSQL, Jupyter, L^AT_EX

PROJECTS

DeepRAG: An Optimized RAG System for Research Document Analysis

Mar. 2026 – May. 2026

University of Maryland, College Park

- Architected an enterprise-grade **Retrieval-Augmented Generation (RAG)** pipeline, processing complex 1,000+ page PDFs via **LlamaParse** to seamlessly preserve highly structured Markdown tables and layouts.
- Replaced rigid token-window splitting with dynamic **Semantic Chunking** (via **Gemma-3**), accurately partitioning documents by contextual shifts to drastically enhance semantic boundaries and retrieval accuracy.
- Optimized vector storage overhead and search latency by **3×** via **Matryoshka Representation Learning (MRL)**, efficiently truncating *nomic-embed-text* dense vectors to 256 dimensions.
- Engineered a robust **Weaviate v4** (Dockerized) vector database supporting **Hybrid Search** (Dense + BM25), augmented by **Multi-Query Expansion** and advanced **Cross-Encoder Reranking**.
- Orchestrated **DeepSeek-R1** (via **Ollama**) as the core reasoning engine, fusing reranked context chunks with high-parameter linguistic reasoning to produce heavily grounded, source-cited responses.
- Validated end-to-end system architecture using an automated **RAGAS evaluation suite**, achieving near-perfect retrieval with **0.98 Context Recall** and **0.93 Context Precision**, alongside robust generative scores of **0.72 Faithfulness** and **0.63 Answer Relevancy**.

Generative AI Clinical Decision Support Pipeline

Apr. 2026 – May 2026

University of Maryland, College Park

- Built an end-to-end **clinical decision support pipeline** processing over **4,000+** MTSamples transcriptions into structured intelligence and actionable insights.
- Integrated **domain-specific BERT architectures** (*bert-base-uncased-clinical-ner*) fine-tuned on **i2b2/n2c2** corpora for **Named Entity Recognition**, precisely extracting medical problems, tests, and administered treatments with **F1 > 0.87**.
- Deployed a **PubMedBERT-based multi-class classifier** achieving **93% accuracy** in routing clinical notes across 13 distinct medical specialties.
- Orchestrated Google's **Gemma3-4B** via local **Ollama** REST APIs, enforcing strict **JSON schemas** to perform advanced clinical reasoning and logically infer comprehensive **treatment recommendations**, evaluating generation fidelity via **ROUGE-L** and **BERTScore**.
- Engineered an interactive **Streamlit** application with proactive **GPU memory management** (VRAM flushing) to sequentially execute multi-stage transformer pipelines on local hardware without bottlenecking.

Algorithmic Marketing Attribution & ROAS Optimization Engine

Apr. 2026 – May 2026

University of Maryland, College Park

- Engineered a unified marketing analytics architecture integrating macro-level **Google Merchandise Store** spend logs with micro-level **Criteo** touchpoint sequences (2.5GB+), mapping daily channel expenditures directly to user conversion paths.
- Architected a highly tuned **Marketing Mix Model (MMM)** via Meta's **Robyn**, leveraging geometric **adstock transformations** and Hill functions to precisely quantify lagged advertising effects, diminishing returns, and organic baseline revenue.
- Developed robust **Multi-Touch Attribution (MTA)** algorithms, benchmarking heuristic rules (time-decay, position-based) against advanced data-driven approaches (**Markov Chains** and **Shapley values**) to properly allocate conversion credit across multi-channel journeys.
- Designed a mathematical **Reconciliation Engine** that fuses top-down MMM budget bounds with bottom-up MTA granular crediting, utilizing non-linear solvers to dynamically redistribute channel spend and maximize global **Marginal ROAS**.
- Validated model precision through rigorous time-series backtesting, continuously evaluating top-down MMM fit via **NRMSE** and **MAPE**, and algorithmic fairness of MTA touchpoint weights against naive last-click baselines.
- Deployed a full-stack interactive **Streamlit** dashboard rendering channel ROI saturation curves, touchpoint transition heatmaps, and actionable, algorithmic budget reallocation strategies for enterprise constraints.

Financial Volatility Modeling & Risk Forecasting using GARCH

Oct. 2025

University of Maryland, College Park

- Developed a validated **GARCH(1,1)** model on **14 years** of daily log returns (2010–2024) to forecast conditional volatility for an equity asset.
- Quantified high volatility persistence ($\alpha_1 + \beta_1 = 0.9736$) via ACF/PACF analysis of squared returns and ARCH-LM testing.
- Estimated Long-Run Volatility at $\sigma_{LR} = 1.66\%$ and generated a short-horizon forecast of $\sigma_{t+1} = 1.33\%$, identifying a low-volatility regime with slow mean reversion.
- Validated model specification through standardized residual diagnostics, confirming white-noise errors and correct model fit.

Univariate Time Series Forecasting using ARIMA

Sep. 2025 – Oct. 2025

University of Maryland, College Park

- Applied the **Box-Jenkins methodology** to a publicly available economic time series - including ADF testing for stationarity and ACF/PACF analysis - to identify an optimal **ARIMA(2,1,1)** specification.
- Selected final model using **AIC/BIC** criteria; validated residuals as white noise via the Ljung-Box test and heteroscedasticity checks.
- Achieved measurable improvement in forecast accuracy (RMSE) over baseline historical averaging, demonstrating model generalizability on a held-out test window.

Formula 1 Race Outcome Prediction (1950–2024)

Nov. 2024 – Dec. 2024

University of Maryland, College Park

- Engineered features across **14 relational tables** covering driver stats, constructor records, circuit attributes, and weather conditions for 70+ seasons of F1 data.
- Built and compared classification models (Logistic Regression, Random Forest, Gradient Boosting) to predict race podium finishes, identifying grid position as the single highest-impact feature.
- Quantified accident risk by circuit, revealing **Monaco** and **Spa-Francorchamps** as statistically higher-incident tracks based on historical incident rates.
- Assessed driver/constructor consistency across seasons using longitudinal performance metrics, surfacing key patterns in championship-winning behavior.